

# TrackLink: Enabling Near-Field Beamfocusing and MIMO for Fast Moving LEO Satellites

**Abstract**—Distributed near-field ground arrays can combine commodity panels for dish-class gain and line-of-sight MIMO rank. Low Earth orbit (LEO) motion, however, makes their kilometer-scale focus both extremely narrow and short lived: public ephemeris can point each panel, but cannot maintain carrier-phase coherence across the full aperture, and a measured channel is stale before a later uplink reaches the satellite. Existing localize-then-focus and channel-reuse methods therefore cannot meet the narrow-angle and fast-clock requirements together. We present *TrackLink*, a ground-side system that separates slow analog pointing from fast digital phase maintenance. On the downlink, *TrackLink* anchors the effective per-panel channel with a full-channel pilot and tracks only phase increments with sparse pilots. On the uplink, where no future measurement exists, it factors the channel into a computable geometric term and a measured quasi-static residual, then predicts the phase at satellite arrival. The ground panels share a frequency reference and periodically phase-calibrate their transmit and receive chains; remaining calibration error is included in the residual. To acquire this predictor from a coarse orbital prior, CFFR combines a measured wrapped-phase intercept, an unwrapped geometric increment, and a closed-form fringe-rate correction, avoiding fine satellite localization and global ambiguity search. Satellite-side equalization preserves multiplexing without returning calibration feedback. In simulations using the published STARLINK-1821 orbit, *TrackLink* realizes 12.02 dB of uplink array gain at a 10 ms turnaround, 0.02 dB from perfect channel knowledge, while stale CSI falls to the incoherent floor. CFFR has zero observed catastrophic failures across the tested 1m–30 km prior-error sweep at  $910\times$  fewer operations than exhaustive search, assuming the residual remains quasi-static over the prediction horizon.

**Index Terms**—Near-field beamfocusing, distributed arrays, LEO satellite, line-of-sight MIMO, coherent combining, Moving target, phase tracking, channel aging

## I. INTRODUCTION

Rapidly expanding Low Earth orbit (LEO) constellations are driving unprecedented satellite-to-ground traffic, but the ground station infrastructure has not scaled at the same pace [1]–[4]. A ground station with parabolic dishes provide high gain, but mechanical steering limits each antenna to one active satellite and introduces substantial downtime during handoffs [5], [6]. Monolithic phased arrays enable rapid electronic steering and simultaneous links, but matching dish-class gain requires tens of thousands of antenna elements [6], [7]. An emerging alternative distributes commodity phased-array panels over a kilometer-scale aperture, using roughly one-third as many elements [7]. Although fewer antenna elements offer lower gain, the enlarged aperture places the satellite in the array’s radiative near field and enables multiple parallel LoS-MIMO streams. So even if each MIMO stream is low gain, the existence of multiple LoS-MIMO streams provides the

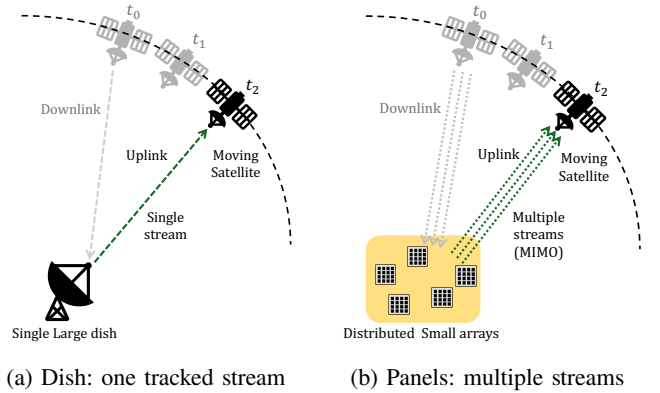


Fig. 1: **Ground-station architectures.** (a) A mechanically steered dish. (b) Distributed phased-array panels. Panel counts are schematic.

high aggregate throughput. Recent work [7] shows feasibility of coherent beamfocusing and LoS-MIMO for satellite links. However, they assume a frozen satellite channel for their analysis, leaving a critical question unanswered: *can these benefits be maintained for a fast-moving LEO satellite?*

A fast-moving satellite requires frequent beam-focusing updates to maintain coherent gain. This raises two questions: how precisely must the array focus in angle, and how often must this focus be updated? First, the kilometer-scale aperture creates an extremely narrow distributed focus. Staying within 3 dB of peak gain requires angular alignment within  $2.26 \times 10^{-4}$  degree (0.8 arcseconds) (See Fig. 4(a)). In comparison, public two-line element (TLE)-based orbit estimates provide only about  $0.104^\circ$  accuracy, which is  $462\times$  too coarse [8]. Second, near the fastest part of a pass, a satellite moving at 7.5 km/s crosses this 3 dB region in only  $285 \mu\text{s}$ . Mobility therefore imposes two simultaneous requirements: (i) arcsecond-scale angular alignment and (ii) sub-millisecond beam-focus updates.

Existing approaches fall into two broad categories, but neither meets these requirements. The first follows a localize-then-focus philosophy, where the satellite’s angle and range are estimated before synthesizing the corresponding beam [9]–[14]. However, these methods cannot reliably achieve the arcsecond-scale angular alignment required by a kilometer-scale distributed array. We also cannot naively scan for the best beam, because the sparse aperture produces many aliased grating lobes with similar responses, making the correct focus ambiguous [15], [16]. The second category estimates the

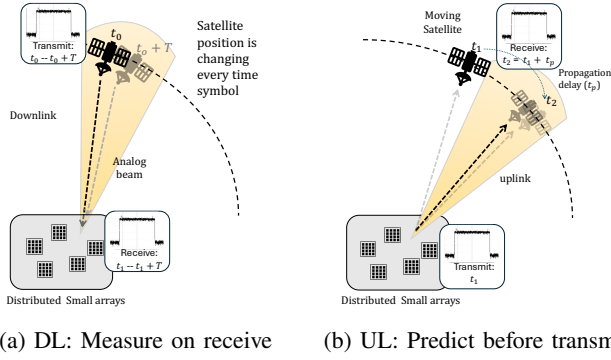


Fig. 2: **Link timing.** (a) Downlink measures current phase; (b) Uplink predicts future phase.

instantaneous channel or predicts its evolution from pilots, Doppler, or motion parameters [17]–[21]. Channel measurement alone cannot close the loop quickly enough when the distributed focus expires in only  $285 \mu\text{s}$ . Existing predictors are primarily designed for conventional far-field or fully digital arrays, and do not provide the carrier-accurate channel prediction required across kilometer-separated analog phased-array panels.

We present *TrackLink*, the first ground-side framework for maintaining near-field beamfocusing and LoS-MIMO over a kilometer-scale distributed array to a fast-moving LEO satellite. Our key insight is rather than localizing the satellite with arcsecond-scale angular accuracy, *TrackLink* uses coarse angle estimates to steer each panel and directly maintains the relative phase across panels. To achieve this, *TrackLink* separates beamfocusing into two timescales: slow analog pointing and fast digital phase alignment. The analog stage steers each panel’s broad local beam using public TLE information and remains valid for seconds. The digital stage aligns phases across the full aperture and updates at a sub-millisecond timescale. This separation removes fine pointing from the analog control loop. The remaining challenge is how to obtain and maintain the relative phases at the required rate.

To address digital phase tracking, another key observation is that phase acquisition differs fundamentally between the two link directions, as illustrated in Figure 2. On the downlink, the required phase is embedded in the received signal and can be observed by the ground array as the signal arrives. On the uplink, however, the array must select its transmit weights in advance for the channel that will exist when the signal reaches the moving satellite. This channel lies about 10 ms into the future, far beyond the  $285 \mu\text{s}$  lifetime of the current focus. Therefore, the downlink can rely on current phase measurements, whereas the uplink must predict the future phase. This makes uplink beamfocusing the more challenging problem.

**Channel aging and residual-offset challenge.** A measured channel captures the true per-panel phase, including hardware, calibration, and propagation effects, but it quickly becomes stale as the satellite moves. Geometry can capture this motion,

but it cannot predict the unknown offsets across panels. *TrackLink* resolves this tension by decomposing the channel as  $\mathbf{h}(t) = \mathbf{h}_{\text{geo}}(t) \odot \mathbf{h}_{\text{res}}(t)$ . The geometric term changes rapidly but can be computed from orbital motion and panel locations. The residual contains effects that geometry cannot model and varies slowly over the short prediction horizon, so it can be measured and carried forward. Neither term is sufficient alone. A held measurement misses the satellite’s motion, while geometry alone misses the panel-dependent offsets. Thus, *TrackLink* measures what geometry cannot provide and computes what motion changes too quickly to measure.

**Wrapped-phase prediction challenge.** Even with this decomposition, predicting the future geometric phase from a coarse orbit prior remains difficult because phase is observed only modulo  $2\pi$ . A natural approach is to search over candidate satellite positions and select the one that best explains the measured phase. However, the sparse kilometer-scale aperture produces many ambiguous solutions, while iterative methods require a much tighter initialization than the public orbit prior provides. We address this challenge with CFFR, a closed-form fringe-rate predictor that directly predicts the future per-panel phase without first localizing the satellite. For a pair of panels, phase changes fastest when their baseline aligns with the satellite’s apparent motion and slowest when it is perpendicular. CFFR separates this evolution into a measured starting phase and a modelled unwrapped change, then uses a closed-form fit across panel pairs to correct the remaining motion error. It therefore predicts the phase required at the future arrival time without fine angle–range localization or a carefully initialized search.

We evaluate whether these insights translate into system-level gains using the published orbit of STARLINK-1821 over a complete pass. At a 10 ms turnaround, *TrackLink* achieves 12.02 dB of uplink array gain, only 0.02 dB below perfect channel knowledge, while a stale channel falls to the incoherent floor. CFFR records zero catastrophic failures across prior errors up to 30 km and requires  $910\times$  fewer operations than exhaustive search. Over the full pass, a four-antenna satellite delivers  $1.84\times$  the data volume of a single-antenna satellite. These results are obtained through simulation under the stated channel model and assume that the measured residual remains quasi-static over the prediction horizon; validating this assumption in hardware is left to future work.

**Contributions.** We make the following contributions:

- **Near-field LEO tracking.** To our knowledge, *TrackLink* is the first framework for maintaining near-field beamfocusing and LoS-MIMO over a kilometer-scale distributed array to a fast-moving LEO satellite.
- **Two-timescale phase maintenance.** *TrackLink* separates slow analog pointing from fast digital alignment, measuring current phase on the downlink and predicting future phase on the uplink.
- **Search-free phase prediction.** We design CFFR, a closed-form predictor that combines measured phase, modelled motion, and cross-panel correction without fine localization or global position search.

- **Reproducible evaluation.** We evaluate complete satellite passes against stale-channel, geometry-only, search-based, and perfect-knowledge baselines, and release our simulator and scripts with the paper.

## II. BACKGROUND AND MOTIVATION

### A. A pass is short, and the satellite will not help

A LEO ground station never owns its endpoint for long. A satellite rises, crosses, and sets in minutes, and capacity is not bought once and pointed once but re-earned on every pass against a target moving at 7.5 km/s. Everything a pass will deliver is delivered inside that window.

A gateway operator serving a commercial constellation also inherits a waveform, a pilot structure, and an antenna configuration that were fixed before launch and stay fixed for the spacecraft's service life. We call this the *frozen fleet* constraint, and it has two consequences that shape every decision in this paper. Knowledge of the satellite is limited to what a public catalog publishes, which supports angular tracking to  $0.104^\circ$  [8], so the ground station cannot ask the satellite where it is. And the satellite runs no calibration loop on our behalf, so an uplink must be precoded before transmission with no cooperation from the far end. The link itself is a Ka feeder pair, 18.5 GHz down and 28.05 GHz up, a ratio of 1.5162, and the interval between measuring the downlink and transmitting on the uplink is tens of milliseconds. The ground panels therefore share a frequency reference and periodically phase-calibrate their transmit and receive chains locally; the remaining mismatch is part of the measured residual channel. All of the intelligence lives on the ground, or it does not exist.

### B. The aperture sets the near-field regime, not the panel

At 28.05 GHz ( $\lambda = 10.695$  mm) the Rayleigh distance  $2D^2/\lambda$  of a  $D = 1$  km ground aperture is roughly  $1.9 \times 10^5$  km, so a satellite at 400 to 1200 km sits deep inside the radiative near field of the array while remaining in the far field of every individual 0.171 m panel. The regime is a property of the aperture, not of the elements that fill it. Its operational content is that a plane-wave model misstates differential path length across the aperture, so focusing weights depend on where the wavefront came from and not only on which direction it arrived from.

### C. Separation creates rank, and curvature does not

Placed side by side, every ground antenna sees an essentially identical channel to every satellite antenna. The channel matrix is rank one to within noise, extra antennas buy diversity and array gain, and the link delivers exactly the single-stream capacity  $\log_2(1 + \text{SNR})$  and no more [22]. Separating the antennas is what opens a second stream, which is the reason the array is spread at all.

The tempting explanation is that spherical curvature over a kilometer of ground manufactures the rank, and it is wrong. Curvature is common-mode across a two-element transmit array and cancels to second order in the difference between the two columns of the channel matrix. What governs rank is the

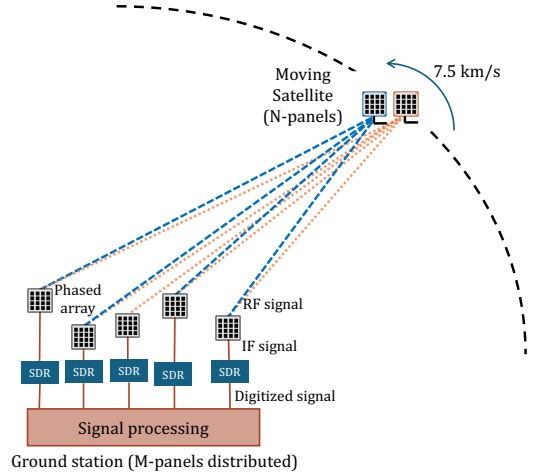


Fig. 3: **TrackLink architecture.** Sixteen independently digitized panels share a frequency reference. Each panel is pointed in analog; their outputs are focused across the full aperture in digital.

paraxial degrees-of-freedom product  $\text{DoF} = D_{\text{rx}}d_{\text{tx}}/(\lambda R)$ , the classical line-of-sight MIMO spacing criterion [23]. What the near field buys instead is *observability*: because range enters differential path length quadratically across the aperture, a kilometer-scale array can recover the shape of the wavefront that reached it. That recovered phase, and never a satellite position, is what Section III focuses and predicts against.

## III. DESIGN

The Introduction reduced mobility to one systems problem: maintain relative phase across a kilometer-scale aperture. This section derives *TrackLink* by first assigning pointing and coherence to different control timescales, then showing why the downlink can measure its current phase while the uplink must predict its future phase. Unless stated otherwise, the steering requirements use the narrower 28.05 GHz uplink beam; Section III-B uses the 18.5 GHz downlink and Section III-C returns to 28.05 GHz.

### A. TrackLink separates pointing from coherence

*TrackLink* distributes 16 commodity  $32 \times 32$  panels across a 1 km by 1 km site (Fig. 3). Each panel has one RF chain and a 0.171 m aperture, and a central processor combines the digitized panel streams. Four panels fix the corners and the others are placed aperiodically. The resulting 1414.2 m diagonal sets the digital resolution, whereas the panel aperture sets the analog field of view. The panels share a frequency reference. *TrackLink* also assumes that the ground panels' transmit and receive chains are periodically phase-calibrated; the remaining calibration error is included in the residual channel defined below.

The two apertures cannot be steered by one loop. Each panel has a  $3.174^\circ$  beam, while coherent combination across the site synthesizes a beam only  $4.51 \times 10^{-4}$  degrees (1.625 arcseconds) wide (Fig. 4a). Public two-line elements are

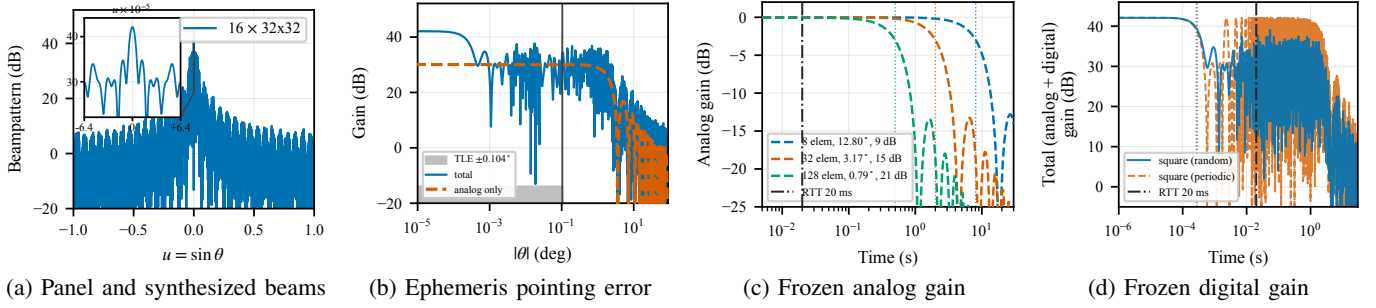


Fig. 4: **The two steering scales.** (a),(b) Angular response of the panel and distributed aperture. (c),(d) Gain after their steering weights are frozen.

accurate to  $0.104^\circ$  [8]. That estimate lies comfortably inside a panel beam, but is  $462\times$  coarser than the digital 3 dB tolerance (Fig. 4b). Motion separates the loops even more clearly: frozen panel steering remains within 3 dB for 2.01 s, whereas the full-aperture focus remains there for only  $285\ \mu\text{s}$  (Fig. 4c,d).

Improving the ephemeris or mechanical tracker cannot close both gaps. It can reduce coarse pointing error, but it does not recover carrier-phase path differences across the site, and no measure-to-steer loop closes in hundreds of microseconds. Searching for the focus is no better: covering the  $0.104^\circ$  prior at the  $2.26 \times 10^{-4}$ -degree tolerance takes 462 beams per axis, and the focus itself moves within  $285\ \mu\text{s}$ , leaving under a microsecond per beam. *TrackLink* therefore assigns each state to the timescale that can maintain it: public ephemeris keeps the satellite inside every panel's broad analog beam, while digital processing maintains relative phase across panel outputs. Fine angle is never an analog control variable.

The fast state is easy to miss in a conventional link budget. A common phase rotation, including common Doppler, multiplies the channel by one scalar and changes neither coherent gain nor rank. A phase rotation that differs across panels changes the combining vector and destroys both. The differential Doppler spread is only 1.484 kHz, roughly one percent of a subcarrier, yet an uncompensated array loses 1 dB in  $197\ \mu\text{s}$ . The state that the digital stage must maintain is therefore not absolute carrier phase, common Doppler, or a fine angle. It is the mean-centered vector of panel phases.

That state has a useful physical decomposition. For panel  $j$  at position  $\mathbf{p}_j$  and satellite position  $\mathbf{q}(t)$ , write

$$\mathbf{h}(t) = \mathbf{h}_{\text{geo}}(t) \odot \mathbf{h}_{\text{res}}(t), \quad h_j^{\text{geo}}(t) = \frac{1}{d_j(t)} e^{-jk d_j(t)}, \quad (1)$$

where  $d_j(t) = \|\mathbf{q}(t) - \mathbf{p}_j\|$  and  $k = 2\pi/\lambda$ . The geometric factor changes rapidly but is computable from motion and surveyed panel positions. The residual contains hardware, cable, remaining calibration, and propagation terms that geometry does not provide, so it must be measured.

Over the sub-50 ms horizon considered here, *TrackLink* models the residual as quasi-static relative to the geometric term:  $\mathbf{h}(t + \tau) \approx \mathbf{h}_{\text{geo}}(t + \tau) \odot \mathbf{c}$ . This is a system assumption, not something a geometry-driven simulator can validate. Section IV-A therefore states that boundary explicitly and tests estimator, calibration, and receiver errors without

claiming hardware validation. Equation (1) also exposes the link asymmetry. On the downlink the current product arrives with the signal and can be measured. On the uplink the product is needed at a future time, so *TrackLink* carries the measured residual forward and computes the geometric change.

## B. Downlink: the received signal supplies the phase

Motion changes phase, not magnitude. Over a 2 ms interval relative magnitudes remain nearly constant while relative phases accumulate by more than  $1,200^\circ$  (Fig. 5a,b). A localize-then-focus receiver would search angle and range until a modeled spherical wavefront matches those phases [9]–[11]. Across a sparse kilometer aperture, wrapped phase creates many look-alike locations, and that search solves for a position the receiver does not need.

On receive, the signal itself supplies the focusing weights. With one RF chain per panel, maximum-ratio combining uses

$$\mathbf{w}_m^* = \frac{h_m^*}{\|\mathbf{h}\|}, \quad \left| \sum_{m=1}^M \mathbf{w}_m^* h_m \right|^2 = \|\mathbf{h}\|^2. \quad (2)$$

No satellite position appears in (2). Coarse analog steering only has to place the signal inside each panel beam; a per-panel channel estimate then realizes the narrow digital focus directly. The same estimate includes the receive-chain phase of each ground panel, so the downlink does not require a separate transmit/receive reciprocity calibration.

The obvious way to keep that estimate current is to send a full channel pilot in every symbol. It would follow the phase, but would spend spectrum repeatedly estimating delay and magnitude that barely change. *TrackLink* instead uses the existing NR division of labor. A front-loaded DM-RS provides a high-energy full-channel anchor once per slot, and PT-RS tracks only the phase increment between anchors using sparse frequency resources. The latest estimate is applied on every symbol even though a full channel is not re-measured every symbol. At the selected  $L = 1, K = 2$  density, PT-RS costs 4.167% of resource elements; its lower energy is sufficient because it estimates an increment rather than delay or a new channel.

Figure 5d reflects that schedule. The downward markers are DM-RS anchors. The larger peaks occur during the two symbols at the start of the next slot, before its front-loaded DM-RS, while the last PT-RS estimate is still being held; they

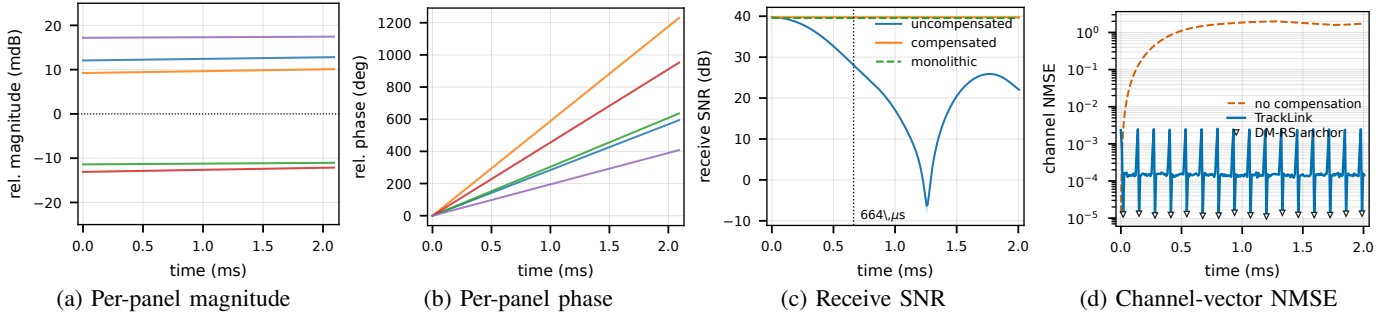


Fig. 5: **Downlink phase maintenance over 2 ms.** (a),(b) Effective per-panel channel. (c) Receive SNR for held, tracked, and monolithic arrays. The dotted rule is one synthetic-beam traversal. (d) Common-phase-invariant channel NMSE.

are not PT-RS failures. Without updates, receive SNR falls by 10 dB within  $609 \mu\text{s}$  and eventually reaches the incoherent floor (Fig. 5c). DM-RS plus PT-RS keeps the worst mean-centered weight error at  $3.38^\circ$ , within the  $8.27^\circ$  focusing budget, and preserves the gain of a monolithic array with the same element count. Denser time sampling is more valuable than denser frequency sampling at fixed overhead because geometric drift, not estimator noise, is the dominant error. The downlink therefore needs neither fine localization nor a feedback loop: it measures current relative phase, refreshes its increment, and combines.

### C. Uplink: predict phase across the turnaround

The uplink must choose its weights before transmission for the channel that will exist at satellite arrival. During a 10 ms turnaround, differential phase advances by  $2,230^\circ$  mean-centered across the array, compared with the  $8.27^\circ$  focusing budget. A held channel loses 1 dB within  $125 \mu\text{s}$  and is already at the incoherent floor at 10 ms. The downlink solution cannot simply be delayed.

Several obvious repairs retain only part of Eq. (1). Reusing the measured phase ignores motion. Scaling that wrapped phase by the uplink-to-downlink carrier ratio also fails because the unobserved integer turns are not scaled. Geometry carries those turns exactly, but geometry alone omits the residual per-panel offsets and also reaches the incoherent floor. The uplink needs both: the residual from the latest measurement and the geometric channel evaluated at the future arrival time.

The other obvious solution is to localize first. Maximum-likelihood search and an extended Kalman filter score candidate positions against wrapped phase. For this aperture the likelihood has 141 local maxima. Both methods work from a tight seed, but a pass begins from a kilometer-scale ephemeris prior spanning hundreds of resolution cells. Search then locks to aliases or becomes intractable. The missing primitive is not a finer global position estimate; it is a search-free estimate of how differential phase is changing.

1) *CFFR: predict phase rate without localization:* Over the short observation and prediction window, differential phase is linear to first order. For panels  $j$  and  $r$ , differentiating their path-length difference gives the classical interferometric fringe rate

$$\frac{d}{dt}(\phi_j - \phi_r) = \frac{2\pi}{\lambda} \omega b_{jr,\parallel}, \quad (3)$$

where  $\omega$  is the apparent angular rate and  $b_{jr,\parallel}$  is the panel-pair separation projected onto the apparent direction of motion. A pair separated perpendicular to the track sees little phase change; a pair aligned with the track sees the largest. Every wavelength of differential path change contributes one phase turn. That rate is what CFFR estimates; recovering it from the measurement is the difficulty.

**Wrapped and discontinuous measurements.** The receiver observes that slope only modulo  $2\pi$ , and only when a downlink pilot burst arrives; between bursts the link carries data, so we treat the pilot period  $T_p$  as a parameter. On the 1 km baseline the phase advances  $459^\circ$  per millisecond, more than a turn per millisecond, so the integer turns accumulated between bursts must come from somewhere other than the samples. Every predictor below sees the same samples and must extrapolate 10 ms past the last one; they differ in where those turns come from and in what they extrapolate.

**Naive unwrapping fails at the first gap.** The textbook tracker unwraps the samples sequentially, placing each on the branch nearest its predecessor, then fits a polynomial and extrapolates it. That choice is correct only while consecutive samples differ by less than half a turn, which at  $459^\circ$  per millisecond requires samples under  $0.39 \text{ ms}$  apart. A 9 ms gap between bursts spans  $4,130^\circ$ , so the unwrap lands on the wrong branch at every gap and fails every trial from  $T_p = 10 \text{ ms}$  onward (Fig. 7).

**Ephemeris turns fail as the gap grows.** The orbit prior can supply the count. The alternative unwraps inside each burst, where samples are dense, places the whole burst on the branch nearest the prior's own phase ( $2\pi n$ , with  $n$  rounded from the burst-averaged discrepancy), and fits the same polynomial. It survives a small gap: a wrong  $n$  that is the same on every burst is a common offset the beamformer never sees. It fails in two ways as the gap grows. First, every burst is a fresh commitment on every baseline, and the commitment is wrong once the prior's phase error drifts past half a turn between bursts; with noise removed, that alone fails 50% of trials at  $T_p = 10 \text{ ms}$ . Second, it still fits raw phase, whose slope of hundreds of degrees per millisecond must be recovered from each 1 ms burst, and it extrapolates the noise on that slope ten times the burst length. Together they fail 27% of trials at  $T_p = 10 \text{ ms}$  and every trial from 50 ms (Fig. 7).

**CFFR: subtract the motion first, then fit what is left.**

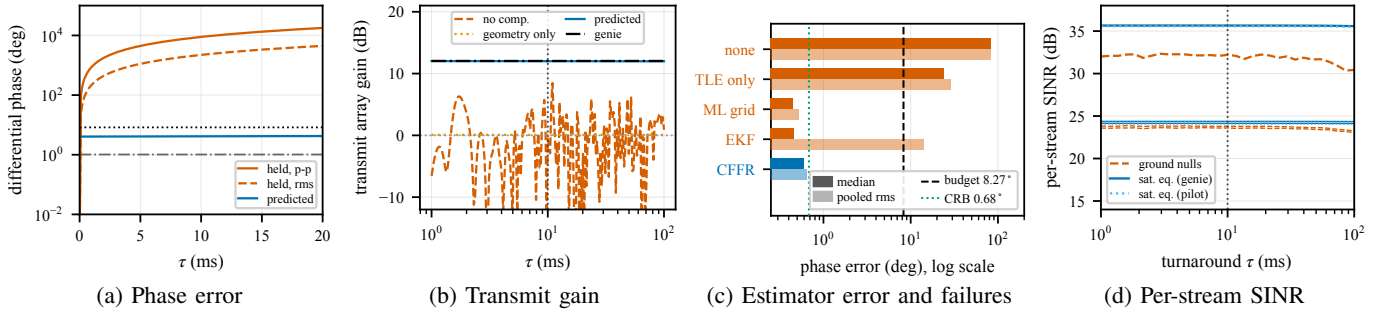


Fig. 6: **Uplink prediction.** Sixteen ground panels over a 1 km square. (a),(b),(d) STARLINK-1821 at 394 km culmination. (c) Estimator comparison at 550 km and a 10 ms turnaround.

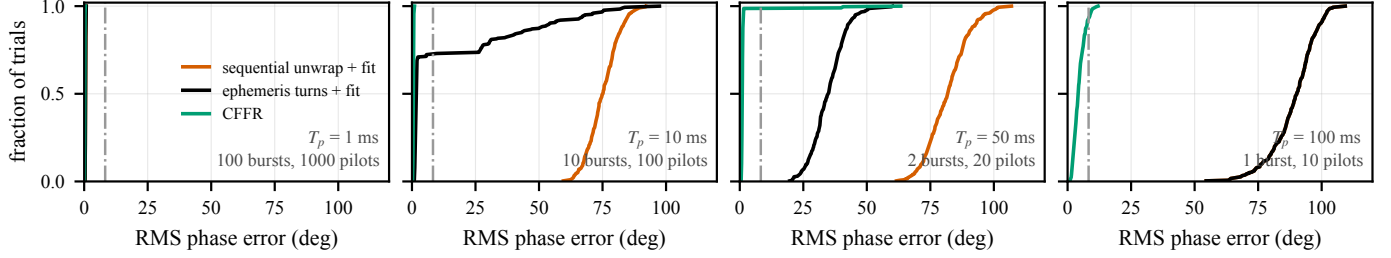


Fig. 7: **Prediction error against pilot period.** CDF of mean-centered RMS phase error at the 10 ms turnaround over 300 trials. A ten-pilot burst repeats every  $T_p$  inside a fixed 100 ms window. The dash-dot rule is the  $8.27^\circ$  focusing budget.

CFFR removes the prior's predicted phase before anything is unwrapped (Fig. 8). Step 1 forms the wrapped differential phase  $z_{jn} = \text{wrap}(\phi_{jn} - \phi_{rn})$ . Step 2 evaluates, from the ephemeris state  $(\mathbf{q}_0, \mathbf{v}_0)$ , each panel's path length  $\hat{d}_j(t) = \|\hat{\mathbf{q}}(t) - \mathbf{p}_j\|$ , the modelled differential phase  $\hat{s}_j = -k(\hat{d}_j - \hat{d}_r)$ , and its increment  $D_{jn} = \hat{s}_j(t_n) - \hat{s}_j(t_0)$ , then subtracts it. Because  $D$  is a difference of path lengths in meters, it carries every full turn. What remains,

$$y_{jn} = \text{unwrap}(\text{wrap}(z_{jn} - D_{jn})), \quad (4)$$

is the only quantity the fit sees; its intercept absorbs the static residual phase modulo  $2\pi$ .

**The residual is small and linear.** Two properties follow from Eq. (3). First,  $y$  moves at the prior's velocity error, about  $1^\circ$  per millisecond in the left thumbnail of Fig. 8 against  $459^\circ$  for the raw phase, so the unwrap in Eq. (4) never faces a branch choice and CFFR never commits an integer. Second,  $y$  is linear in that error. With  $\hat{\mathbf{u}}_j$  the unit vector from panel  $j$  toward the prior position and  $G_j = -k(\hat{\mathbf{u}}_j - \hat{\mathbf{u}}_r)$ ,

$$y_j(t) = a_j + (\mathbf{G} \delta \mathbf{v})_j t, \quad (5)$$

with one intercept  $a_j$  per baseline and one state  $\mathbf{x} = (\delta \mathbf{v}, \delta \mathbf{a})$  shared by all of them, where the acceleration error enters through  $t^2/2$  (Step 3). Fifteen baselines therefore constrain six unknowns, regularized toward the prior along weak directions, and CFFR solves that least-squares system in closed form; the right thumbnail plots the fitted slopes against the measured ones with  $R^2 = 0.98$ . Sharing one state also matters when bursts are sparse: the shared solve averages over 15 baselines the slope noise that a per-baseline polynomial would extrapolate, and holds it to the prior. Step 4 evaluates the fitted line at  $t + \tau$ , and Step 5 adds back  $D_j(\tau)$ , evaluated from

the prior at the arrival time rather than extrapolated, to return  $\hat{\phi}_j(\tau)$ . The fringe-rate relation is classical [20], [24], [25]; to our knowledge, turning raw wrapped per-panel phase into a future channel through a measured intercept, an unwrapped geometric increment, and a closed-form correction is new.

**CFFR holds where both baselines fail.** Figure 7 sweeps  $T_p$  over a fixed 100 ms window at the 10 ms turnaround; a ten-pilot burst repeats every  $T_p$ , so a longer period also means fewer pilots. At  $T_p = 1$  ms the stream is effectively continuous and all three estimators agree within  $0.6^\circ$ . From 10 ms on, CFFR holds medians of  $0.48$ ,  $0.99$  and  $4.18^\circ$  with failure rates of 0, 1.3 and 7.3%, where sequential unwrapping fails every trial and ephemeris turns fail 27%, 100% and 100%. Across the 49 ms gap its residual changes by tens of degrees in a typical draw, inside the half turn its unwrap tolerates; the rare draw that exceeds it is the 1.3%. At  $T_p = 100$  ms a single burst leaves both baselines a free per-baseline polynomial through ten samples spanning 1 ms, and both reach  $90^\circ$ ; CFFR extrapolates the same samples through one shared, regularized state and holds  $4.18^\circ$ .

Because CFFR never forms a position likelihood, it has no ambiguity peak on which to initialize. The fit observes transverse motion; radial range remains largely at its prior, which is acceptable because the output is panel phase, not a reported satellite position. Planarity is important: horizontal panel pairs keep altitude error in the weak direction of the fit. The resulting deployment conditions are explicit: the site must be approximately coplanar, the panels must share a frequency reference and periodic TX/RX phase calibration, and the closed-form estimator is intended for links above roughly 5 dB. Section IV-C evaluates prior width, failure probability, and computation cost.

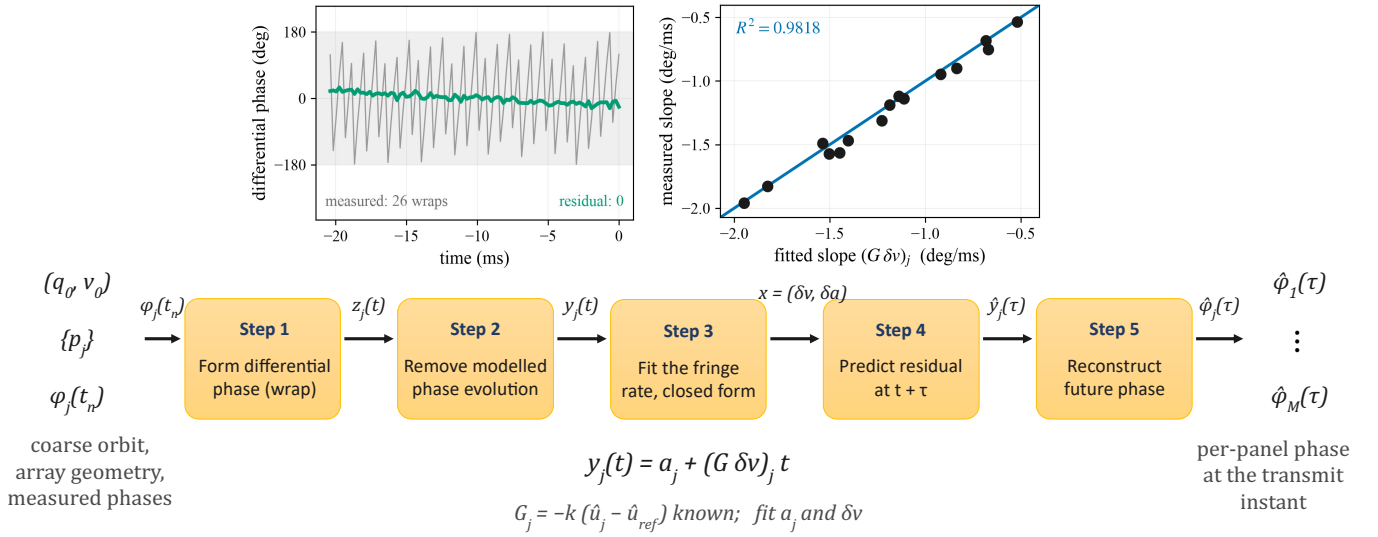


Fig. 8: **CFFR**. Five steps from wrapped per-panel phase and a coarse orbit to per-panel phase at the transmit instant. Left thumbnail: measured differential phase and its residual after Step 2. Right thumbnail: fitted against measured fringe rate over the 15 baselines.

There is one hardware asymmetry that receive-side measurement does not remove. A downlink pilot traverses each panel's receive chain, whereas the uplink traverses its transmit chain. Their differential phase must therefore be known before precoding. The periodic ground-side calibration assumed in Section III-A supplies this transmit/receive phase difference, and its remaining error is part of  $\mathbf{h}_{res}$ . This is a local array-calibration requirement; *TrackLink* does not require calibration feedback from the satellite.

2) *Satellite equalization turns nulling into focusing*: Uplink multiplexing appears to require the ground to preserve spatial nulls more tightly than it preserves coherent power. That is true only if the satellite uses the receiver implied by the ground's predicted channel. Let  $\mathbf{G} = \mathbf{U}\Sigma\mathbf{V}^H$  and let diagonal  $\mathbf{D}$  contain the ground's remaining panel-phase error. The effective channel contains  $\mathbf{V}^H\mathbf{D}^H\mathbf{V}$ . Its first-order generator is Hermitian, so this factor is unitary to second order and does not change singular values to first order. A satellite that estimates and equalizes the effective channel therefore turns a null-placement requirement into the same power-delivery requirement used for focusing. This is ordinary receiver processing based on uplink pilots; it does not send calibration information back to the ground. In the evaluated two-stream link, the 0.1 dB phase tolerance increases from  $0.67^\circ$  to  $8.06^\circ$ , and pilot-based equalization retains most of that margin.

The two links now close for different reasons. The downlink measures the effective channel after its receive-chain offsets have acted. The uplink starts from ground-calibrated transmit chains, carries the measured residual forward, and uses CFFR to compute the geometric change. Section IV-B tests whether that computed future channel beats a stale measurement, while Section IV-D tests whether the resulting gain and rank survive a complete pass.

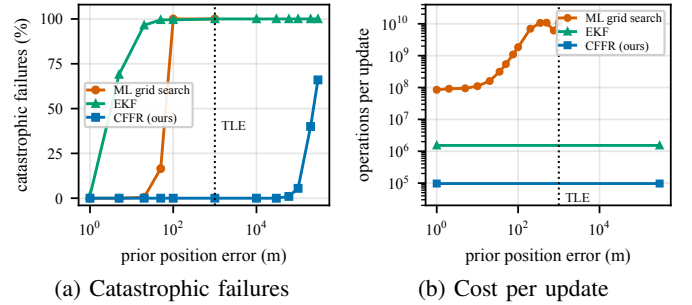


Fig. 9: **Cold-prior acquisition**. Failure probability and per-update operations versus prior position error. The dotted rule marks the 1 km TLE prior; the ML curve stops there because larger grids exceed memory.

#### IV. EVALUATION

The evaluation follows the dependency chain in the design. We first define a common physical model, metric set, and collection of fair baselines. We then ask four questions. Does a channel computed for satellite arrival beat the measured channel it replaces? Does that prediction preserve multiplexing as well as coherent gain? Can CFFR start from the coarse prior available at the beginning of a pass? Finally, do those component results survive a complete pass rather than only one favorable instant? We close by stating what the simulator cannot validate and where the design fails.

##### A. Methodology and common setup

**A shared geometry and link model.** Satellite trajectories are propagated from published two-line elements with SGP4 through *skyfield*. Unless stated otherwise, we use STARLINK-1821 (NORAD 46718) over a mid-latitude site. The pass reaches  $87.3^\circ$  elevation and 394 km slant range at culmination. For every ground panel and satellite antenna, the simulator computes the exact spherical path length; no

TABLE I: Common parameters and system assumptions.

| Parameter                       | Value                                                           |
|---------------------------------|-----------------------------------------------------------------|
| Carrier, downlink / uplink      | 18.5 / 28.05 GHz                                                |
| Numerology                      | 120 kHz SCS, 1024-FFT,<br>122.88 MSa/s                          |
| Ground panels                   | $16 \times 32 \times 32$ at $\lambda_{UL}/2$                    |
| Panel synchronization           | Shared frequency reference;<br>periodic TX/RX phase calibration |
| Panel side; uplink<br>beamwidth | 0.171 m ( $16\lambda_{UL}$ ); $3.174^\circ$                     |
| Ground array extent             | 1 km $\times$ 1 km (1,414 m diagonal)                           |
| Slant range; elevation mask     | 394 to 2,285 km; $10^\circ$                                     |
| Satellite antennas              | 1, 2, or 4                                                      |
| Propagation; turnaround         | 2.63 ms two way; 10 ms budget                                   |

plane-wave or paraxial channel approximation is used. Individual experiments then add the effects they test: panel response, additive noise, phase-estimation error, calibration and coordinate perturbations, OFDM timing, precoding, and receiver processing. Table I lists the parameters and system assumptions shared across those experiments.

**Metrics expose both typical behavior and failures.** For focusing, we report mean-centered per-panel phase error and realized array gain relative to one panel; the coherent ceiling for 16 panels is 12.04 dB. Mean-centering removes common phase, which changes neither gain nor rank. For MIMO, we report the worst-stream SINR after regularized zero-forcing and receiver equalization. For acquisition, a trial is catastrophic when its predicted phase error exceeds the  $8.27^\circ$  focusing budget; every estimator is therefore reported with a median, pooled RMS error, and failure rate. For a complete pass, we integrate throughput into delivered volume rather than reporting only peak rank or peak SNR.

**Comparisons differ in one source of information at a time.** The stale arm applies a measured channel after the turnaround. TLE-only uses public orbit data with no phase measurement. Geometry-only uses the exact arrival geometry but omits the measured residual. *TrackLink* combines the latest measured residual with geometry at arrival, and genie uses the exact arrival channel. Acquisition additionally compares CFFR with maximum-likelihood grid search, an extended Kalman filter, and local geometric fits. Paired arms use identical orbits, noise draws, total transmit power, pilot observations, and receiver processing; no method silently falls back to another after failure.

**The simulator does not validate residual physics.** Its propagation channel is geometry plus additive noise, and its injected per-panel residual is held over each trial. Thus the decomposition  $\mathbf{h} = \mathbf{h}_{\text{geo}} \odot \mathbf{h}_{\text{res}}$  holds by construction. The experiments test the consequences of the model and errors in measurement, prediction, calibration, and receiver processing. They do not establish that a real atmospheric and hardware residual remains quasi-static; that remains the explicit assumption stated in Sec. III-A.

## B. Prediction closes the uplink loop

**A causal comparison.** Sixteen panels precode one stream at the 10 ms turnaround. The stale arm asks whether a recent measurement remains usable. Geometry-only asks whether a better ephemeris can replace channel measurement. *TrackLink* and genie isolate the remaining prediction error. All four arms use the same ground-side frequency reference and periodic TX/RX phase calibration; remaining calibration error is included in the measured residual.

**The measured channel is not merely degraded; it is the wrong channel.** At 10 ms, motion has advanced the mean-centered differential phase by  $2,230^\circ$ , while *TrackLink* leaves  $4.13^\circ$  (Fig. 6a). The stale channel loses 1 dB within  $125 \mu\text{s}$  and realizes  $-0.57$  dB at the turnaround, already at the incoherent floor. *TrackLink* realizes 12.02 dB against a 12.04 dB genie, a 0.02 dB gap (Fig. 6b). The impact of prediction is therefore the entire 16-panel coherent gain, not a small refinement to feedback.

**Both channel factors are necessary.** Geometry-only is given the exact satellite position at arrival and still realizes only  $+0.08$  dB. Its failure is the controlled ablation for the residual: orbit products contain path change, but not the static cable, hardware, calibration, and propagation offsets between panels. Conversely, stale CSI contains those offsets but not the geometric change. Only their combination approaches genie. This is why improving the TLE cannot replace *TrackLink* and why faster pilot feedback cannot replace prediction.

**A measurement also cannot cross the carrier split.** Reusing the measured 18.5 GHz downlink phase at 28.05 GHz yields 0.15 dB of uplink gain. The obvious correction, scaling wrapped phase by the 1.5162 carrier ratio, yields  $-0.03$  dB. Both have approximately  $90^\circ$  mean-centered weight error. Scaling fails because a measurement reveals phase only modulo  $2\pi$ ; it cannot recover the omitted turns. Geometry carries path length in meters, so scaling the geometric term is exact and yields 12.03 dB. The duplex gap is therefore free to geometry and impassable to a wrapped measurement.

**Multiplexing closes at the receiver.** Ground-side nulling alone tolerates only  $0.67^\circ$  of per-panel phase error before the worst of two streams loses 0.1 dB. When the satellite estimates and equalizes its effective channel from uplink pilots, that tolerance rises to  $5.28^\circ$ ; with genie channel knowledge it is  $8.06^\circ$ , a  $12.1\times$  relaxation (Fig. 6d). The single-stream arm is the control: all receivers agree because no spatial null exists to move. For two and three streams, equalization converts a first-order null-placement error into the second-order power loss derived in Sec. III-C2. Above 15 dB reference SNR, each usable stream reaches the same modulation-and-coding ceiling; relative to an equal-element monolithic aperture, two and three streams therefore deliver  $2\times$  and  $3\times$  spectral efficiency. Satellite equalization uses ordinary uplink pilots and returns no calibration information to the ground.

## C. CFFR acquires from a cold prior

We next test cold-start acquisition from the kilometer-scale prior available at the beginning of a pass. Each method

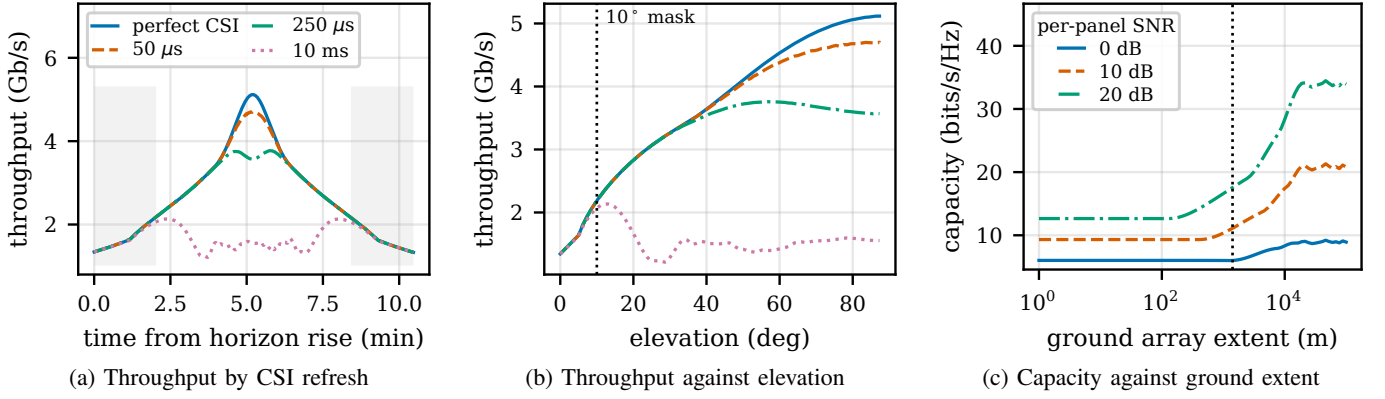


Fig. 10: **One STARLINK-1821 pass.** The first two panels sweep stale-CSI refresh. The third sweeps ground extent; the dotted rule marks the 1 km design.

receives 96 noisy phase snapshots at 20 dB and predicts 10 ms ahead. Prior-width points use 50–200 trials; the EKF sweep uses independent draws under the same model and failure threshold. One transverse resolution cell is 4.16 m.

**A median can hide alias lock.** At a 1 m prior, maximum-likelihood search and CFFR have 0.44° and 0.58° median errors and neither fails. The EKF has a similar 0.46° median, yet fails in 2.5% of trials and reaches 14.3° pooled RMS error (Fig. 6c). We therefore report failure rate alongside typical error.

**Prior width separates CFFR from seeded methods.** In the independent EKF sweep, failures reach 69% at a 5 m prior and nearly 100% by 20 m. Grid search first fails at 20 m, fails every trial by 100 m, and exceeds memory beyond 1 km. CFFR has zero observed failures through 30 km because its measured intercept absorbs the starting phase while geometry carries every full turn (Fig. 9a). Its first observed failures at 60 km, rising to 66% at 300 km, bound rather than weaken that claim.

**Search-free acquisition fixes computation.** CFFR uses  $9.7 \times 10^4$  operations per update independent of prior width. The EKF costs  $1.5 \times 10^6$  operations, 16× more, yet loses its phase branch as the prior widens. Grid search grows from  $8.8 \times 10^7$  operations at 1 m to  $1.1 \times 10^{10}$  at the 1 km TLE prior, or roughly  $10^3$ – $10^5 \times$  CFFR (Fig. 9b). CFFR trades slight median accuracy under a tight prior for sub-degree error and fixed cost across the operational range.

#### D. A complete pass, not one favorable instant

The final experiment propagates the real 628 s STARLINK-1821 pass from horizon to horizon and scores the 382 s above a 10° elevation mask. Each satellite antenna transmits 30 dBW over 480 MHz into a 200 K receiver. Throughput is integrated over time while the channel refresh interval and satellite aperture are varied.

**Refresh rate determines delivered data.** Perfect CSI delivers 160 GB above the elevation mask. Refreshing at 50  $\mu$ s preserves 98.3% of that volume, 250  $\mu$ s preserves 92.4%, and a 10 ms held channel preserves only 49.6% (Fig. 10a). The pass therefore turns the sub-millisecond phase dynamics from

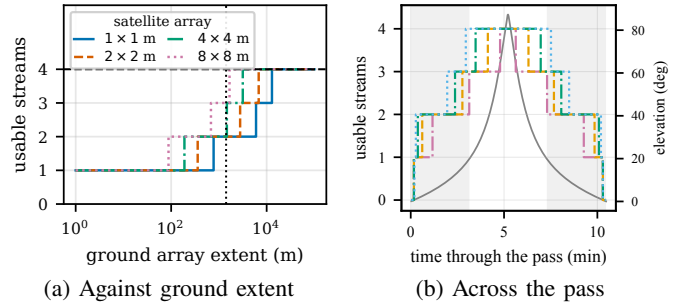


Fig. 11: **Usable streams against satellite aperture.** Four satellite antennas are retained in every arm; only their spacing changes.

Sec. III-A into a direct system outcome rather than only a beamforming tolerance.

**The best-SNR instant is the worst stale-CSI instant.** Near culmination, range is shortest and received SNR is highest, but transverse angular motion and differential phase rate also peak. The 250  $\mu$ s curve consequently develops a local throughput minimum at 86–87° elevation (Fig. 10b). A link budget based only on peak SNR would provision the least tracking exactly where the channel changes fastest.

**Rank is set by aperture at both ends.** With four satellite antennas on the 1 m baseline assumed by the design, four streams are available for only 8.0% of the pass and rank two is available for 77.4%. Widening the satellite baseline to 8 m makes four streams available for 43.6% of the pass and raises pass-integrated throughput from 1.84× to 2.77× that of one satellite antenna (Fig. 11b). The number of satellite RF chains is unchanged; only spatial observability changes.

At the 1 km ground design point, the same four-antenna satellite supports two streams with a 1 m satellite baseline and three with an 8 m baseline (Fig. 11a). Ground extent and satellite baseline trade approximately one for one through the paraxial factor  $D_{rx}d_{tx}/(\lambda R)$ . Arrangement also matters: holding satellite extent fixed but changing a linear array to a 2 × 2 planar array can move the design point by one stream. A peak rank without the satellite aperture and pass fraction that produced it is therefore not a capacity result.

**Static Distributed Near-field Beamfocusing:** Representative works in near-field communications investigate beam focusing and channel estimation once a user enters the radiative near field of an extremely large aperture array, based on the spherical-wave propagation model [9], [10], [12]–[14], [26]–[34]. These works typically first estimate the user’s angle and distance to perform localization, and then focus the beam onto the recovered position [9]–[12], [14]. A representative work is the modular-array design by the Björnson et al, which applies the MUSIC algorithm independently on each sub-array to obtain the angle and then recovers distance via geometric triangulation [11], [35].

However, these techniques are only effective for meter-scale apertures, fully digital arrays, and quasi-static users. When applied to a kilometer-scale distributed satellite ground station that must deliver both high gain and line-of-sight MIMO rank [7], the synthesized beam contracts to arcsecond scales and localization accuracy requirements rise by orders of magnitude. Existing near-field methods cannot meet this precision, and analog-digital hybrid panels cannot supply the per-antenna observations they require. Conventional localize-then-focus procedures therefore become inapplicable and fail to provide the per-panel phase needed for coherent combining.

**LEO Channel Prediction and Localization:** Prior studies have established that instantaneous CSI ages quickly under LEO motion and must be complemented by orbital geometry or prediction [17], [36]–[39], [39], [40]. Subsequent approaches employ statistical CSI for multi-satellite precoding [18], [41]–[43] or learn to predict path parameters [19], [44]. Public ephemerides are also used to determine satellite direction [8], [45]. These methods operate under far-field assumptions. They produce angle, direction, or path-parameter estimates rather than per-panel phase, and do not address phase wrapping. On a large aperture, where path differences span many wavelengths, such unresolved  $2\pi$  wraps shift panels by full cycles and turn coherent combining into cancellation.

**Fringe-Rate-Based Phase Tracking:** The classical fringe rate of radio interferometry is the rate of differential phase change across a baseline and is used to recover source and observer motion [20], [24], [25]. The closest system-level realization is StarLoc [21], which combines interferometric angle estimation with orbital modeling on a 13 cm three-antenna array to localize Starlink transmitters and reports an angular error of approximately  $0.7^\circ$ . These methods consume frequency observables on compact far-field arrays and return kinematic quantities such as angle or velocity. They do not produce the per-panel channel phase required on a large near-field aperture, where the same angular accuracy is orders of magnitude too coarse to focus.

Prior static feasibility work has established this distributed near-field architecture for a stationary satellite [7]. The present work addresses phase acquisition under motion by tracking per-panel phase rather than estimating satellite position.

*TrackLink* shows that mobility in a kilometer-scale distributed LEO array is not fundamentally a fine-pointing problem; it is a differential-phase maintenance problem with two timescales. Separating slow panel steering from fast digital coherence lets public ephemeris close the analog loop, while the two links close differently. The downlink directly measures the current per-panel phase from the received signal. The uplink has no future measurement, so it predicts the phase needed when the signal reaches the satellite. CFFR acquires that predictor without fine localization or a global ambiguity search, and satellite-side equalization preserves multiplexing without returning calibration feedback. Under the stated simulation model, this separation recovers 12.02 dB of transmit array gain at a 10 ms turnaround, within 0.02 dB of perfect channel knowledge, while stale CSI reaches the incoherent floor. The remaining empirical question is whether real atmospheric and hardware residuals remain quasi-static over the prediction horizon. The broader lesson is to measure what geometry cannot provide and compute what motion changes too quickly to measure. The channel is stale; geometry is not.

## VII. USE OF AI DISCLOSURE

OpenAI Codex was used to refine grammar, generate code blocks, and assist with plotting scripts. The authors reviewed all AI-assisted outputs and verified the generated plots against simulation results; they retain responsibility for the methodology and conclusions.

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